

speeds of up to 120MB/sec (800X) and writing speeds up to 30MB/sec (200X).

SanDisk and Samsung are other popular brands offering SSDs. Laptops, such as the Sony VAIO, are also beginning to adopt SSDs instead of the traditional hard disks.

SSDs are still much more expensive than traditional storage, and hence not very popular at the personal usage level. However, they are beginning to play an interesting role in enterprise storage systems. Mahadevan explains, "A significant development in the storage space is the addition of a tier 0 of storage composed of SSDs and other flash products to complement the performance-optimised disk drives and capacity-optimised disk drives."

Safely locked

Consumers and vendors are paying a lot of attention to the protection of data. While security at the enterprise level is obvious, it is interesting to note the security features available at the portable storage level.

ThinkPad's portable secure hard drive was showcased at last year's Consumer Electronics Show as one of the best innovations in the computer peripherals category. The hard drive is optimised for safeguarding essential data while on-the-move and offers high-level 128-bit Advanced Encryption Standard (AES) security within a slim frame.

Companies are realising that thumb drives are a huge security threat, and therefore incorporating more security features. For example, in October, Lexar announced its new JumpDrive Safe S3000 FIPS, which it claims is the world's first tamper-resistant smart card-based Federal Information Processing Standard (FIPS) 140-2 level 3 validated flash drive.

Software and hardware get inseparably wedded

The face of storage has changed. "If you study the evolution, it had been hardware driving the software, till the nineties. But, today, it is the soft-

ware which is driving most innovations. There are multiple software in storage... in fact, I would call it a framework of multiple software accomplishing the storage task for different needs," says Vish Sivaswamy, managing director, Conjoin Group.

Actually, storage has become an inseparable combination of hardware and software. P.K. Gupta, director and chief architect-information protection practice, EMC Global Services, endorses this by saying, "Storage no longer means storing data only. It has evolved to play a wider role of information management by bringing together storage, security, intelligence, virtualisation and automation."

Without software it would be impossible to address today's needs of security, data availability, scalability, etc. "Technology evolutions like data deduplication, advanced replication technologies, disk and tape encryption are some of the examples where software has played a major role in revolutionising storage," says Subram Natarajan, executive IT consultant, IBM Systems and Technology Group.

Bharat Kumar, vice president-sales & marketing, Asia Pacific, Japan and Middle East, Tandberg Data, clarifies, "Software has made data warehousing and archiving an easier task. It has also made storage scalable, sharable, enabling it to be split between users."

Susumu Kobayashi, country manager, Buffalo India, points out another key area of software. "Storage operating systems such as Linux-based ones and Windows Storage Server contribute to expanding the use of network-attached storage by adding new features," he says.

Deduplication

Data deduplication can be called as one of the most promising recently introduced technologies. It comprehensively addresses all the capacity problems in conjunction with performance, security and management

issues. For example, IBM's Deduplication Gateway solutions help companies to simplify and accelerate information protection, help lower operational cost, end energy usage, and manage more data with simpler infrastructure.

"Data deduplication can reduce the cost associated with storing large amounts of data. Deduplication is the art of intelligently reducing storage needs through elimination of redundant data so that only one instance of a data set is actually stored," explains Sandeep K. Dutta, vice president-storage, systems and technology group, IBM India/SA.

The next step to deduplication is global deduplication. SearchStorage.com, in its acclaimed annual predictions, cites, "Global deduplication, managing islands of dedupe and virtual tape library (VTL) appliances and sharing dedupe data among them, is a much-needed innovation for the next-generation dedupe products."

Thin provisioning

"Modern data centres are constantly asked to do more with less, especially during times of tightening IT budgets. Thin provisioning enables them to deliver exactly that. While storage technologies like data deduplication and compression are also very effective

